NEWTON SCHOOL

PRE-CALCULUS | HOMEWORK · 30 PROBLEMS

Mastery Set: Log Properties → Polynomial Theorems

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Format: Each section opens with 2 worked examples (study these first), followed by practice problems. Show all work.

Part 1 — Logarithm Properties (12 problems)

Goal: *Master* $\log_b(b) = 1$, $\log_b(1) = 0$, *product rule, quotient rule, power rule, and change of base.*

Worked Example 1.A — Evaluating a Logarithm

Evaluate $\log_4(64)$.

Strategy: Ask — "4 to what power gives 64?"

$4^3 = 64$, so $\log_4(64) = 3$.

Equivalently, set $\log_4(64) = x \Rightarrow 4^x = 64 = 4^3 \Rightarrow x = 3$.

$\boxed{\log_4(64) = 3}$

Worked Example 1.B — Expanding with Log Laws

Expand $\log\left(\dfrac{x\,y^2}{\sqrt{z}}\right)$ using log properties.

Step 1. Quotient rule: $\log(x\,y^2) - \log(\sqrt{z})$.

Step 2. Product rule on numerator: $\log(x) + \log(y^2) - \log(z^{1/2})$.

Step 3. Power rule on each: $\log(x) + 2\log(y) - \frac{1}{2}\log(z)$.

$\boxed{\log(x) + 2\log(y) - \frac{1}{2}\log(z)}$

Problems 1–7: *Evaluate each. (No calculator for #1–5.)*

1. $\log_3(81) =$

2. $\log_2\left(\frac{1}{16}\right) =$

3. $\log_5(125) =$

4. $\ln(e^7) =$

5. $\log(10{,}000) =$

6. $\log_4(8) + \log_4(2) =$

7. $\log_3(54) - \log_3(2) =$

Problems 8–10: *Expand fully using log laws.*

8. Expand: $\log_2(x^2 y^3)$

9. Expand: $\ln\left(\sqrt{\frac{a}{b}}\right)$

10. Expand: $\log_5\left(\frac{x^3 y}{z^2}\right)$

Problems 11–12: *Condense into a single logarithm.*

11. Condense: $4\log(x) + 2\log(y) - 3\log(z)$

12. Condense: $\frac{1}{2}\ln(a) - 3\ln(b)$

Part 2 — Solving Log Equations (4 problems)

Always: *state domain first, then solve, then reject extraneous.*

Worked Example 2.A — Single Log

Solve $\log_3(x+1) = 2$.

Domain: $x + 1 > 0 \Rightarrow x > -1$.

Convert to exponential form: $x + 1 = 3^2 = 9 \Rightarrow x = 8$. $8 > -1$ ✓.

$\boxed{x = 8}$

Worked Example 2.B — Sum of Logs

Solve $\log_2(x) + \log_2(x-2) = 3$.

Domain: $x > 0$ and $x > 2 \Rightarrow x > 2$.

Combine: $\log_2\big(x(x-2)\big) = 3 \Rightarrow x(x-2) = 8$.

$x^2 - 2x - 8 = 0 \Rightarrow (x-4)(x+2) = 0 \Rightarrow x = 4$ or -2.

Reject -2 (outside domain). Keep $x = 4$.

$\boxed{x = 4}$

13. Solve: $\log_2(x) = 5$

14. Solve: $\log_3(x-2) = 4$

15. Solve: $\log_2(x) + \log_2(x+2) = 3$ (state domain)

16. Solve: $\log_5(x+1) - \log_5(x-3) = 1$ (state domain)

Part 3 — Exponential Modeling (4 problems)

Tools: *$A = Pe^{rt}$, $A(t) = A_0 e^{kt}$, and the identity $a^t = e^{t \ln a}$.*

Worked Example 3.A — Continuous Compound Doubling

\$1,000 is invested at 5\% compounded continuously. How long until it doubles?

$2000 = 1000\, e^{0.05\, t} \Rightarrow 2 = e^{0.05\, t}$.

$\ln 2 = 0.05\, t \Rightarrow t = \dfrac{\ln 2}{0.05} \approx 13.86$ years.

$\boxed{t \approx 13.86 \text{ years}}$

Worked Example 3.B — Solve $a^x = N$

Solve $2^x = 10$ for x (4 decimals).

Take ln on both sides: $x \ln 2 = \ln 10$.

$x = \dfrac{\ln 10}{\ln 2} = \dfrac{2.30259}{0.69315} \approx 3.3219$.

$\boxed{x \approx 3.3219}$

17. Solve for x (4 decimals): $5^x = 14$. Show change-of-base.

18. A principal of \$3,000 is invested at an annual rate of 5.5%, **compounded continuously**. Using $A(t) = Pe^{rt}$, determine how many years until the investment grows to \$9,000.

19. Convert exponential bases. Rewrite $f(t) = 800\,(1.045)^t$ in the form $f(t) = 800\,e^{rt}$.

(a) Find the exact value of r.

(b) Interpret r as a continuous growth rate (percentage to 2 decimal places).

20. A sample starts at 100 grams and decays to 10 grams after 50 years following $A(t) = A_0 e^{kt}$

.

(a) Find k (exact, in terms of ln).

(b) Find the half-life (years).

Part 4 — Fundamental Theorem of Algebra (3 problems)

Conjugate Root Theorem: *if real coefficients and $a + bi$ is a zero, so is $a - bi$.*

Worked Example 4.A — Find All Zeros Given One Complex Zero

Given i is a zero of $P(x) = x^3 - x^2 + x - 1$, find all zeros.

> Conjugate $-i$ is also a zero. Quadratic factor: $(x - i)(x + i) = x^2 + 1$.
>
> Divide: $\dfrac{x^3 - x^2 + x - 1}{x^2 + 1} = x - 1$ (synthetic or long division).
>
> Third zero: $x = 1$. All zeros: $\{1,\ i,\ -i\}$.
>
> $\boxed{P(x) = (x - 1)(x - i)(x + i)}$

Worked Example 4.B — Build a Polynomial

Build degree 3 with real integer coefficients having zeros 2 and $1 + i$.

> Conjugate $1 - i$ must also be a zero. $\big(x - (1 + i)\big)\big(x - (1 - i)\big) = (x - 1)^2 + 1 = x^2 - 2x + 2$.
>
> $P(x) = (x - 2)(x^2 - 2x + 2) = x^3 - 4x^2 + 6x - 4$.
>
> $\boxed{P(x) = x^3 - 4x^2 + 6x - 4}$

21. Given that $(3 - i)$ is a zero of $P(x) = x^4 - 9x^3 + 30x^2 - 42x + 20$, find **all four zeros** and write $P(x)$ in fully factored linear form.

22. Find a polynomial $P(x)$ of degree 4 with **real integer coefficients** having zeros $x = 2$ (multiplicity 2) and $x = 2 + i$. Expand to standard form $ax^4 + bx^3 + cx^2 + dx + e$.

23. Build a polynomial $P(x)$ of degree 3 with real coefficients having zeros $x = -1$ and $x = 3i$. Expand to standard form.

Part 5 — Synthetic Division & Factor Theorem (4 problems)

Remainder Theorem: *when* $P(x)$ *is divided by* $(x - c)$*, the remainder is* $P(c)$. **Factor Theorem:** $(x - c)$ *is a factor* $\iff$ $P(c) = 0$.

Worked Example 5.A — Synthetic Division

Divide $(x^3 + 2x^2 - 5x - 6) \div (x - 2)$ using synthetic division.

$c = 2$. Coefficients: $1, 2, -5, -6$.

$$\begin{array}{c|cccc} 2 & 1 & 2 & -5 & -6 \\ & & 2 & 8 & 6 \\ \hline & 1 & 4 & 3 & \boxed{0} \end{array}$$

Quotient: $x^2 + 4x + 3$; remainder 0. So $(x - 2)$ is a factor.

$$\boxed{Q(x) = x^2 + 4x + 3, \quad R = 0}$$

Worked Example 5.B — Find Parameter via Factor Theorem

Find k so that $(x - 1)$ is a factor of $P(x) = x^3 + kx - 3$.

By Factor Theorem, $P(1) = 0$: $1 + k - 3 = 0 \Rightarrow k = 2$.

$$\boxed{k = 2}$$

24. Use synthetic division to compute $(2x^3 + 3x^2 - 8x + 3) \div (x - 1)$. State $Q(x)$ and R. Is $(x - 1)$ a factor?

25. Use the Remainder Theorem to find $P(-2)$ for $P(x) = x^4 - 3x^2 + 2x - 5$ using synthetic division.

26. Find k so that $(x - 3)$ is a factor of $P(x) = x^3 - 2x^2 + kx - 6$. Justify with Factor Theorem.

27. Factor completely: $P(x) = x^3 - 7x + 6$. (*Hint:* verify $x = 1$ is a zero, then use synthetic division.)

Part 6 — Polynomial Graphing (3 problems)

Multiplicity rules: *odd* > 1 *= slide / flat-cross, odd* $= 1$ *= cross, even = bounce.*

Worked Example 6.A — Sketching from Factored Form

Analyze $P(x) = (x-1)^2(x+2)$: degree, end behavior, zeros & multiplicity, y-intercept.

> Degree $= 2 + 1 = 3$. Leading term x^3 (positive odd) $\Rightarrow$ end behavior: ↓↑ (left down, right up).
>
> Zeros: $x = 1$ (mult 2, bounce); $x = -2$ (mult 1, cross).
>
> y-intercept: $P(0) = (-1)^2(2) = 2$. Point $(0, 2)$.
>
> **Deg 3, ↓↑; 1 (bounce), -2 (cross); $(0, 2)$**

Worked Example 6.B — End Behavior from Leading Term

State the end behavior of $P(x) = -2x^4 + 3x^3 - x + 5$.

> Leading term $-2x^4$: degree even, coefficient negative $\Rightarrow$ both ends point down.
>
> As $x \to \pm\infty$, $P(x) \to -\infty$. Arrows: ↓↓.
>
> **↓↓ (both ends to $-\infty$)**

28. For $P(x) = (x-1)(x+2)^2(x-3)$: state degree, end behavior, all zeros with multiplicity (classify each as bounce/cross/slide), and the y-intercept.

29. For $P(x) = -x^3(x-2)(x+4)^2$: state degree, end behavior, all zeros with multiplicity (classify each), and the y-intercept.

30. Sketch a polynomial $P(x)$ of **degree 4** satisfying ALL of: (i) zero at $x = -2$ (cross), (ii) zero at $x = 0$ (bounce), (iii) zero at $x = 3$ (cross), (iv) negative leading coefficient. Provide a possible factored equation and a rough sketch.